



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
FULL SYLLABUS
07-08-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH

PAPER-2

Time : 2:30 Hours

Maximum Marks : 210

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **20** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

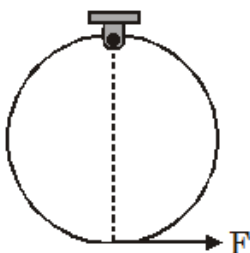
PART-1 : PHYSICS
SECTION-II : (Maximum Marks: 40)

- This section contains **TEN (10)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

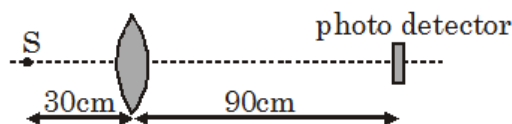
Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. A block is placed on a long and wide inclined plane that makes angle θ with the horizontal. The coefficient of kinetic friction between the block and the plane is $\mu_k > \tan \theta$. After a quick push, the block acquires velocity v_0 that makes angle α with the line of fastest descent. Find the time interval t during which the block is in motion. (Take : $\mu_k = 1$, $\theta = 30^\circ$, $\alpha = 30^\circ$, $v_0 = 5\sqrt{3}$ m/s, $g = 10$ m/s²)
2. A ring of mass $m = 8$ kg and radius $r = 2.5$ m is hinged at its periphery so that it can rotate freely about a vertical axis in a horizontal plane. A force of 72 N is applied so that it is always perpendicular to the diameter. What is the kinetic energy (in J) of the ring after 1 sec?

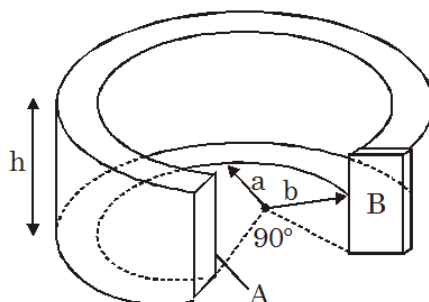


3. A disc shaped photoelectric detector of area 0.5 cm^2 generates current when light of frequency $>$ that of yellow light falls on it. The photoelectron generation efficiency is 20%. In the arrangement shown, S is an isotropic 100 watt point monochromatic light source with $\lambda = 4000 \text{ \AA}$. The focal of the lens is 20 cm and its circular area is 4 cm^2 . Find the photocurrent (in mA) in the detector assuming all rays to be paraxial w.r.t. lens.

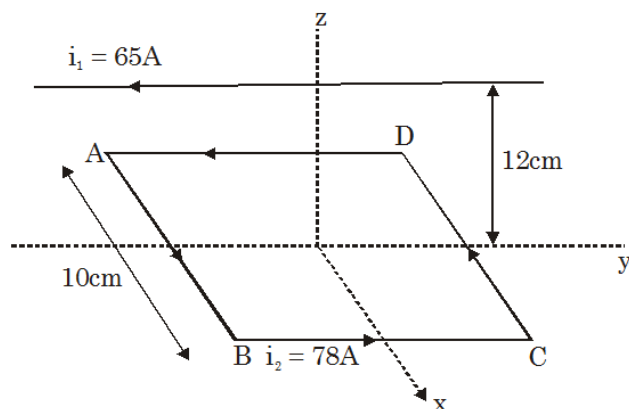


Space for Rough Work

4. Figure shows a hollow cylinder made of copper of height $h = 1 \text{ cm}$, inner radius a and outer radius $b (= 2a)$ with its quarter part removed. Find the resistance (in $\mu\Omega$) of remaining part between faces A and B. Resistivity of copper is $1.68 \times 10^{-8} \Omega\cdot\text{m}$.



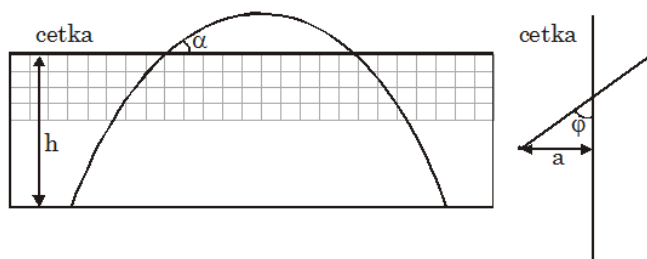
5. Figure shows a square loop 10 cm on each side in the x-y plane with its centre at the origin. An infinite wire is at $z = 12 \text{ cm}$ above y-axis. What is torque (in $\mu\text{N}\cdot\text{m}$) on loop due to magnetic force?



6. In a stationary wave pattern that forms as a result of reflection of waves from an obstacle the ratio of the amplitude at an antinode and a node is $\beta = 1.5$. What percentage of the energy passes across the obstacle ?

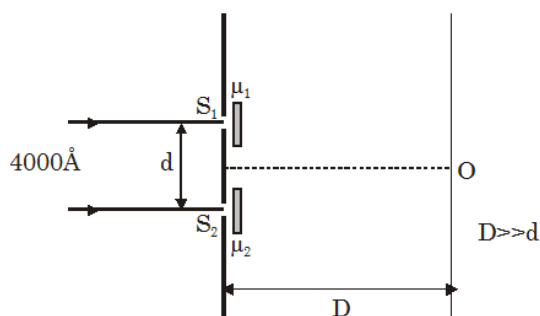
Space for Rough Work

7. When playing volleyball, the player hit the ball close to the ground. The first figure shows the projection of the ball trajectory onto the vertical plane of the net. The tangent to this projection makes an angle $\alpha = 30^\circ$ with the top line of the grid at the point of intersection with it.



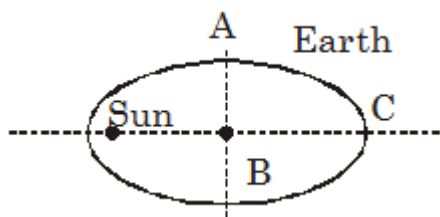
The second figure shows a top view: at the moment of service. At that instant, the player was at a distance $a = 3.5$ m from the net, and the plane of the trajectory forms an angle $\phi = 60^\circ$ with the net. It is known that the speed of the ball immediately after the impact was directed at an angle $\theta = \tan^{-1}(1.2)$ to the horizontal. At what height (in m) above the ground does the trajectory of the ball cross the plane of the net? Net height = 2.4 m. Consider the ball as a point mass and neglect the air resistance.

8. P_1, P_2 are transparent plates having equal thickness $20 \mu\text{m}$ each and refractive indices $\mu_1 = 1.6, \mu_2 = 1.5$. P_1 transmits 75% whereas P_2 transmits 50% of incident energy. Without P_1 and P_2 intensity at O is 4 W/m^2 . Find intensity at O (in W/m^2) after placing P_1 and P_2 . S_1, S_2 are identical slits.

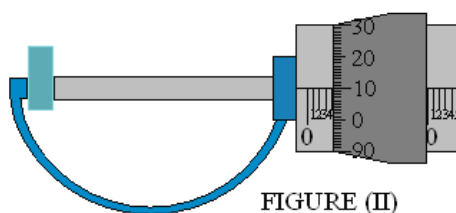
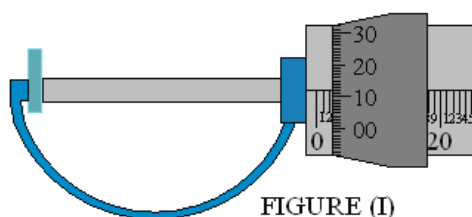


Space for Rough Work

9. Earth revolves around Sun with time period of ' $T \approx 365$ days'. In the figure shown below, find the time taken (in days) by the Earth to move from A to B in the path ACB if the eccentricity of ellipse is 0.5.



10. A screw gauge has some zero error, but its value is unknown. We have two identical slip gauges. When the first one is inserted in the screw, the state of the instrument is shown by diagram (I). When both the gauges are inserted together in series then the state is shown by the diagram (II). What is the magnitude of zero error (in mm) of the instrument? $1 \text{ msd} = 100 \text{ csd} = 1 \text{ mm}$



Space for Rough Work

SECTION-III : (Maximum Marks: 30)

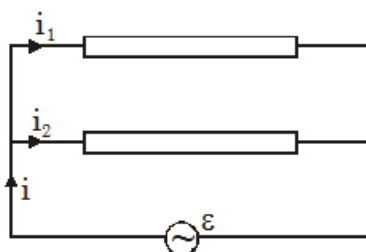
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- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct answer is chosen.

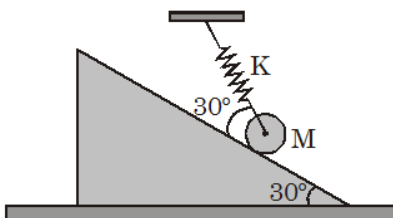
Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. In the circuit shown, a current $i = 20 \sin(100\pi t)$ A flows through an AC source of emf $\varepsilon = 220\sqrt{2} \sin(100\pi t)$ V. In the first branch, current i_1 is $\pi/3$ ahead of i . In the second branch, current i_2 is $\pi/6$ behind i . Find the impedance Z_1 of upper branch in Ω . Write $\left(\frac{Z_1}{2\sqrt{2}} - 5\right)$ as your answer.



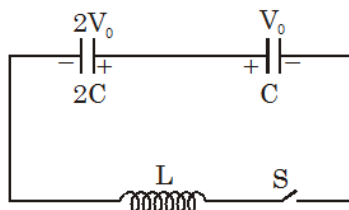
2. A sphere of mass M and radius R is on a smooth fixed inclined plane in equilibrium as shown in the figure. If now the sphere is displaced through a small distance along the plane, what will be the angular frequency (in radian/sec) of the resulting SHM? (Given $K = 4M/3$).



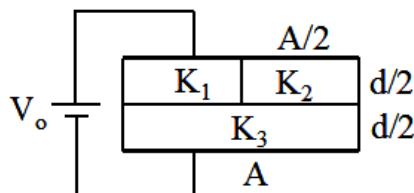
3. The fundamental frequency of a sonometer wire of length $\ell = 1$ m is $f_0 = 200$ Hz. A bridge is now introduced at a distance of $\Delta\ell = 1$ cm from the centre of the wire ($\Delta\ell \ll \ell$). Find approximately, the number of beats (y) heard if both sides of the bridges are set into vibration in their fundamental modes. Fill $\frac{y}{2}$.

Space for Rough Work

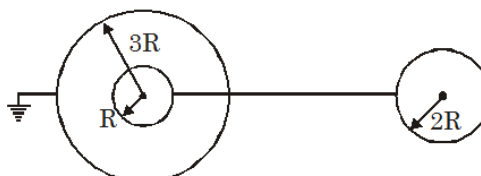
4. Two ideal charged capacitors $2C$ & $C = 3$ Farad are connected with an ideal inductor of inductance L through a switch S . Initially potential difference across $2C$ is $2V_0$ and across C is $V_0 = 4$ Volts. Also, there is no energy stored in the inductor $L = 2$ Henry. After the switch S is closed find the maximum current (in A) flowing in the circuit.



5. An ideal parallel plate capacitor of area A is filled with three dielectric slabs having dielectric constants $K_1 = 3.0$, $K_2 = 5.0$ and $K_3 = 2.0$ and thickness $d/2$ each as shown. The breakdown electric fields in the mediums I, II and III are in the ratio $1 : 2 : 3$. As the applied voltage V_0 is increased, dielectric breakdown occurs first in one of the mediums for $V_0 = 5$ Volts. Write the value of Electric field (in V/mm) in the dielectric medium for which and wherein breakdown occurs first. Take $d = 2$ mm

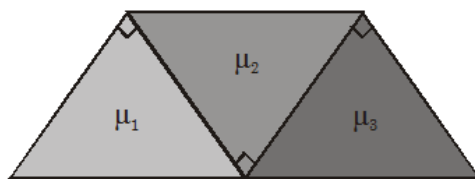


6. Two metal balls of radii R and $2R$ are connected to a long thin conducting wire and has total charge $Q = 21\mu\text{C}$. Then the ball of radius R is placed inside a grounded metal sphere of radius $3R$ as shown. What charge (in μC) will flow through the thin wire in this process ?

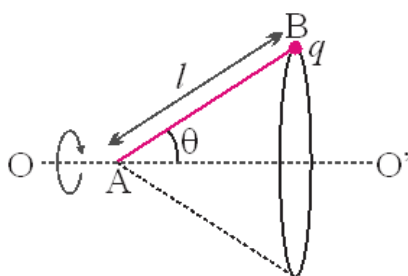


Space for Rough Work

7. Three right angled prisms of refractive indices μ_1 , μ_2 and μ_3 are joined together so that the faces of the middle prism in are in contact each with one of the outside prisms. If the ray passes through the composite block undeviated, find the value of $(\mu_3)^2$. Take $\mu_1 = \sqrt{2}$, $\mu_2 = \sqrt{3}$.

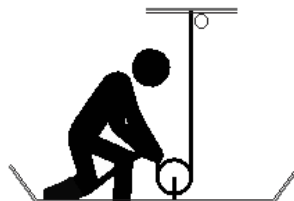


8. A non-conducting rod AB is hinged at end A to a horizontal axis OO' so that it can rotate freely about axis OO' and AB always makes an angle $\theta = 60^\circ$ with axis OO' (i.e. rod AB rotates on a conical surface.) A charge $+q = 2 \mu\text{C}$ is placed on another end B. If rod is rotated with an angular velocity $\omega = 5 \text{ rad/s}$, then calculate magnetic induction (in pT) at point A. Take $\ell = 0.25 \text{ m}$

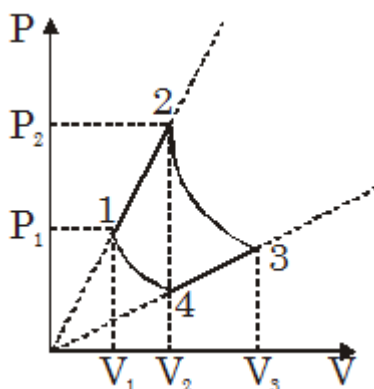


Space for Rough Work

9. A man of mass 55 kg is standing on an open platform of mass 20 kg. The platform is connected to the ceiling through a pulley and the cable going around it as shown. One end of the cable is fixed at O while the other end is being pulled up by the man as shown. The cable and the pulley can be assumed massless. The man pulls with a force of 800 N on the rope. Determine total work done 'W' on the system during an interval from $t = 0$ to $t = 2$ sec. Assuming that the system starts from rest at $t = 0$. Write $3W/100$ as your answer.



10. One mole of a monatomic gas is involved in the cyclic process as shown in the PV graph. Extension of the line segments 1-2 and 3-4 pass through the origin, and the curves 1-4 and 2-3 are isotherms. Find the ratio $\frac{V_3}{V_2}$ if $\frac{V_2}{V_1} = 2$



Space for Rough Work

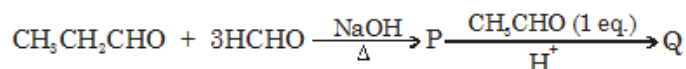
PART-2 : CHEMISTRY
SECTION-II : (Maximum Marks: 40)

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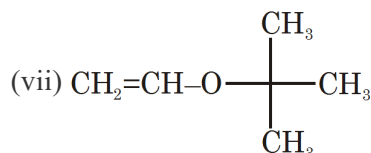
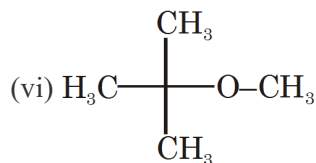
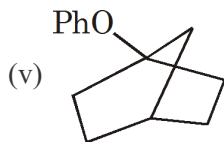
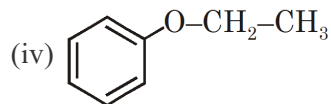
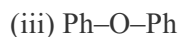
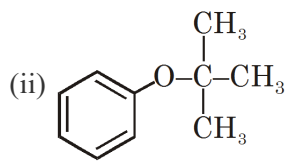
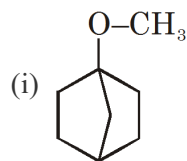
Zero Marks : 0 In all other cases.

1. Find sum of the oxidation number of Fe in given substance :-
Heamatite,
Limonite,
Chromite,
Sodium nitroprusside,
Slage form in metallurgy of Cu,
Mohr's salt,
Product of reaction between Fe + dil. HCl
2. Ortho-phosphoric acid; phosphorus acid and ortho-boric acid all three acid have same number of hydrogens. Find sum of the number of lone pairs in the anions of all the above three oxyacid by considering the reaction with excess NaOH in water.
3. As per C.F.T. Cl-Mn-Cl angle in $[\text{MnCl}_4]^{2-}$ is "θ" what is the angle between e-orbital of central metal and the ligand is –
(If your answer is 84°32' then you have to fill 84.32 in OMR)
4. Formation of (P) is known as Tollen's reaction. Identify total number of stereoisomers for 'Q'

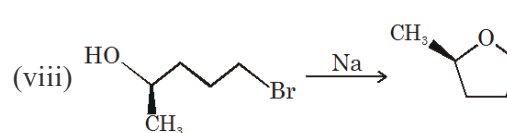
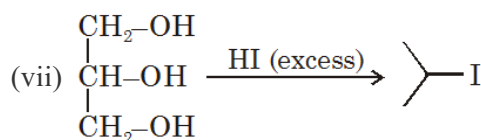
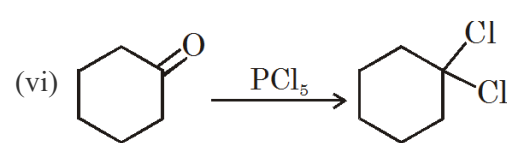
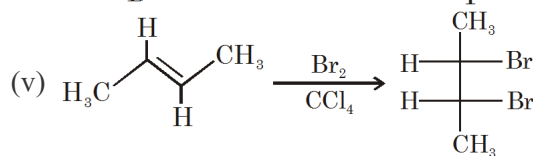
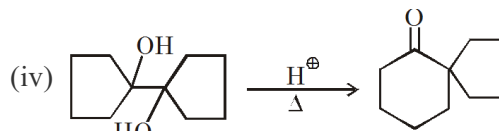
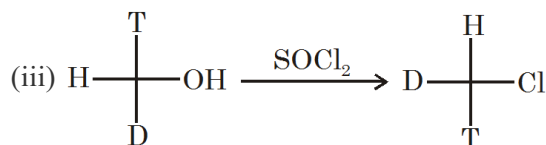
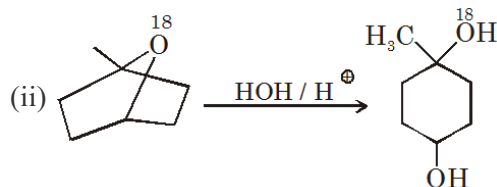
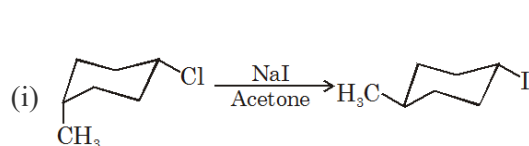


Space for Rough Work

5. Which of the following ether can not be synthesized by Williamson ether synthesis in good yield.



6. In how many reactions, major product is correctly written :



Space for Rough Work

7. If 1 litre water has 162 mg of dissolved calcium bicarbonate, 73 mg of dissolved magnesium bicarbonate and 9.5 mg of dissolved magnesium chloride then permanent hardness of this sample is. (in terms of ppm of CaCO_3)
8. For a certain reaction rate constant k is observed as

$$\ln k = 10^2 + 10^{-2} \ln \left(\frac{T}{K} \right) - \frac{2K}{T}$$
 Find E_a for the reaction in (J/mole) at 300 K
 ($R = 8.3 \text{ J/mole-K}$)
9. EMF of the cell

$$\text{Cu} | \underset{(0.01\text{M})}{\text{Cu}^{2+}} || \underset{(0.1\text{M})}{\text{Ag}^+} | \text{Ag} \text{ at } 25^\circ\text{C}$$
 in volt is -
 (Given : $2.303 \frac{RT}{F} = 0.06\text{V}$, $E_{\text{Cu}^{2+}/\text{Cu}}^0 = 0.34\text{V}$, $E_{\text{Ag}^+/\text{Ag}}^0 = 0.80\text{V}$)
10. For a vanderwaal gas $P_C = \frac{40}{\pi} \text{ atm}$ and $T_C = 200\text{K}$, hence radius of gas molecule in picometer is.

$$\left(R = 0.08 \frac{\ell - \text{atm}}{\text{mol} - \text{K}}, N_A = 6 \times 10^{23} \right)$$

Space for Rough Work

SECTION-III : (Maximum Marks: 30)

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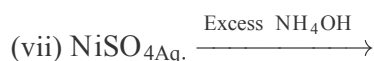
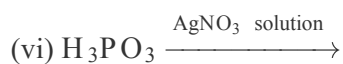
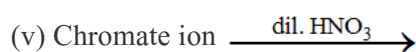
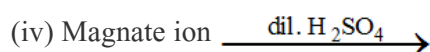
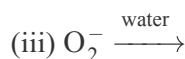
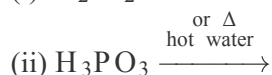
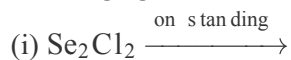
1. For the estimation of % N, 2.8 gm of an organic compound was digested by Kjeldahl's method and evolved NH_3 was absorbed in 120 ml 0.1 N H_2SO_4 . Excess acid was back titrated against 1N NaOH, 2 mL of alkali was required to reach end point. Mass % of N in the organic compound is.
2. Find magnitude of difference in standard Gibb's energy of formation of $\text{H}_2\text{O}(\text{g})$ & $\text{H}_2\text{O}(\text{l})$ at 300K in kJ/mol if Vapour pressure of H_2O at 300 K is 30 torr. [Use : 1 bar = 75 cm Hg]
[Give your answer as nearest integer]
3. If NiS crystallises in rock salt structure where each S^{--} goes to fcc lattice points. If unit cell has formula $\text{Ni}_{0.75} \text{S}_{1.00}$. Then number of effective Ni^{+3} ions per unit cell is.
4. Find the total number of stereo isomers (optical and geometrical isomers) for $[\text{Co}(\text{en})_3][\text{Co}(\text{NO}_2)_2(\text{C}_2\text{O}_4)_2]$

Space for Rough Work

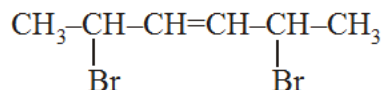
5. Out of given find number of molecular orbitals of homonuclear diatomic molecule, which posses atleast one nodal plane that oriented completely on internuclear axis :-



6. Identify number of reaction(s) in which species given as a reactant act as oxidising agent as well as reducing agent in that reaction :-

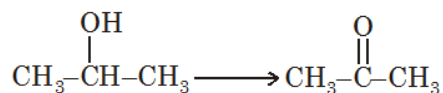


7. Total number of stereoisomer possible by given compound is :



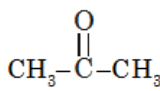
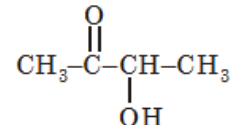
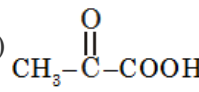
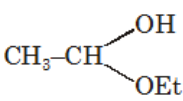
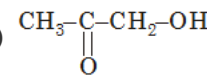
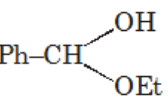
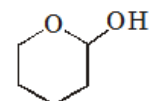
Space for Rough Work

8. How many of reagents can carry out following transformation :



- (a) Cu, 300°C
 (b) O₃ / Zn, H₂O
 (c) NBS
 (d) Zn-Hg / HCl
 (e) Jones reagent
 (f) KMnO₄, NaOH, Δ
 (g) Al(OCMe₃)₃, 
 (h) Collin's reagent
 (i) (i) TsCl, (ii) DMSO, NaHCO₃, Δ
 (j) C₅H₅CrO₃Cl⁻ (PCC)

9. How many of the following will give positive tollen's as well as fehling test ?

- (a) HCOOH
 (b) CH₃-CHO
 (c) 
 (d) Ph-CHO
 (e) 
 (f) 
 (g) 
 (h) 
 (i) 
 (j) 

10. Consider all possible carbonyl compounds including stereoisomers of molecular formula C₅H₁₀O, All these isomers are independently reacted with NaCN / HCN (NOTE : stereoisomers are also reacted separately). The total number of carbonyl compounds that give a racemic product(s) is/are :

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-II : (Maximum Marks: 40)

- This section contains **TEN (10)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Let an ellipse $E_1 : \frac{x^2}{64} + \frac{y^2}{48} = 1$ and two parabolas P_1 and P_2 both passing through ends of minor axis of E_1 . Vertex of each of two parabolas lie on major axis of E_1 . If $x - 2y + 16 = 0$ is common tangent of all the three curves and A be the area bounded by P_1 and P_2 then $\frac{A}{\sqrt{3}}$ is.
2. Let $f(x)$ be a differentiable function satisfying $x(f(x)+1) = x^2 + \frac{x}{e} + \int_0^{x^2} f\left(\frac{t}{x}\right) dt \forall x \in \mathbb{R}_0$.
Let $f'(1) = 1$. If curves $y = f(x)$, $y = \lfloor nx^2 \rfloor$ intersects at (x_n, y_n) for some $n = 1, 2, 3, 4, \dots$ where $y_1 < y_2 < y_3 < y_4, \dots$, then value of $[y_1] + [y_2] + n$ is ($[.]$ denotes greatest integer function)
3. If $7(A + A^{-1}) = \begin{bmatrix} 12 & 25 & 21 \\ 20 & 23 & 77 \\ -14 & 21 & 21 \end{bmatrix}$ and $\det(A) = -7$, then positive square root of value of determinant of matrix $(A^{-1} - I_3)^2$ is

Space for Rough Work

4. If the solution of differential equation $yx^2 dy - y^2 x dx = e^{y^{-3}} dy$ is $kx^2 = 2y^2 + Cy^2$; (where C is integration constant) then the value of k is
5. An ellipse is inscribed in the circle $x^2 + y^2 = 9$ whose major axis is $3x - 4y = 0$. From Point (0, 3) a perpendicular is drawn to the line $3x - 4y = 0$ intersecting the ellipse at P and line at Q such that $PQ = 1$. If F is the distance between both the foci of ellipse then $\frac{144F^2}{119}$ is.
6. If $\vec{p}$ and $\vec{q}$ are orthogonal unit vectors and $\vec{s}$ is a vector such that $\vec{s} \times \vec{q} = \vec{p} - \vec{s}$ then value of $|\vec{s} + \vec{p} \times \vec{q}|$ is
7. Let $f(x) = 4x^4 - \alpha x^3$ and $h(x)$ be a quadratic polynomial satisfying $h(0) + 6 = h'(0) - \beta = h''(0) + 2\gamma = 0$ where $\alpha, \beta, \gamma \in \mathbb{R}$. If $y = f(x)$ and $y = h(x)$ intersect at 4 distinct points with non negative abscissae x_i , $i = 1, 2, 3, 4$ such that $\sum_{i=1}^4 \frac{i}{x_i} = 8$, then value of α is

Space for Rough Work

8. Consider a 5-digit natural number $N = \underline{a} \underline{b} \underline{c} \underline{d} \underline{e}$. Let P be the set consisting of all such N such that each succeeding digit cannot be less than its predecessor and Q be the set of all such N such that each succeeding digit cannot be greater than its predecessor. Then find $n(P \cup Q)$ $\{n(X)$ represents no. of elements of set $X\}$.
9. Harish has 6 fair coins. He tosses all these coins and discards coins which show tail. He then tosses the remaining coins and again discards coins which show tail. Now he tosses the remaining coins for the third time and again discards the coin showing tail. The probability that after this Harish is left with atmost two coins is $\frac{(2401) \lambda}{8^6}$, then determine λ .
10. Let $y = f(x)$ be a function satisfying $x = [y] + \{y\}^3$, $[.]$ denotes greatest integer function and $\{.\}$ denotes fractional part function, then find $\sum_{i=0}^9 Lf'(i)$ (where $Lf'(a)$ is left hand derivative of $f(x)$ at $x = a$)

Space for Rough Work

SECTION-III : (Maximum Marks: 30)

- This section contains **TEN (10)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct answer is chosen.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. Let α and β are the roots of $x^2 - x + 1 = 0$ if $P_k = 2\alpha^k + 3\beta^k$, $k \in \mathbb{N}$ then value of $\frac{\sum_{k=0}^{6n-1} |P_{12k+2} - P_{6k+1}|}{n \left| \sum_{k=0}^{6n-2} (P_{k+2} + P_k) \right|}$ is
(where $n \in \mathbb{N}$).
2. Let $f(x) = (1+x)\ln(1+x) - x - \frac{x^2}{4}$, $\forall x > -1$ and $f'(x) > 0 \forall x \in (0, \alpha)$, then value of $[\alpha]$ is (where $[\cdot]$ denotes greatest integer function)
3. Let two functions $f: \left[\frac{\pi}{2}, \frac{3\pi}{2}\right] \rightarrow [-1, 1]$, $f(x) = \sin x$ and $g: [\pi, 2\pi] \rightarrow [-1, 1]$, $g(x) = \cos x$. If $\sin I(x)$ and $\cos I(x)$ are inverse functions of $f(x)$ and $g(x)$ respectively, then number of solutions of equation $\cos I(x) - \sin I(x) = \cos I(x\sqrt{3})$ is
4. If $\langle a, b, c \rangle$ are direction cosines of line of intersection of three planes $x + y - z = 1$, $(m-1)x + (3m-7)y - 5z = -3$ and $(m-2)x + (2m-5)y = -4$ for some $m \in \mathbb{R}$, then value of $\frac{a+b}{c}$ is

Space for Rough Work

5. Let S be the circle $x^2 + y^2 - 2x - 4y - 4 = 0$ and M be any point on S. Let S_1 be the set of all circles whose center is $R(4 + \cos\theta, 6 + \sin\theta) \forall \theta \in \mathbb{R}$. MP and MQ are tangents to S_1 . Let L and ℓ be the maximum and minimum values of length of diameter of circumcircle of triangle MPQ then $\frac{L}{\ell}$ is
6. Let a vector $\vec{V}_1 = \vec{OP}$, where O is origin and P is (2, 0, 2). Vector $\vec{V}_1$ is rotated about origin by an angle of $\frac{\pi}{3}$ in a plane which is perpendicular to XOZ plane to get vector $\vec{V}_2$ where $\vec{V}_2 \cdot \hat{j} > 0$. Again $\vec{V}_1$ is rotated about origin by right angle in XOZ plane to get vector $\vec{V}_3$. If V is volume of tetrahedron whose coterminal edges are $\vec{V}_1, \vec{V}_2$ and $\vec{V}_3$ then value of $\frac{3V}{\sqrt{6}}$ is.
7. Let $f(x) = \sin^2 x e^{\cot x}$, $x \in (0, \pi)$ then number values of α satisfying $f'(\alpha) = 0 = f''(\alpha)$ is
8. Consider a complex number $z = \frac{\sqrt{5} + 1}{4} + i \left(\frac{\sqrt{5} - 1}{2} \right) \cos \frac{\pi}{10}$ and a set $S = \{z_p | z_p = z^p, p = 0, 1, 2, 3, \dots, 9\}$
 Number of ordered pairs (z_i, z_j) where $z_i, z_j \in S$ such that $\arg(z_i) - \arg(z_j) = \frac{2\pi}{5}$
 (where $\arg z \in [0, 2\pi)$) is

Space for Rough Work

9. Quantities of red and white balls in two urns are shown in table

	Red	White
Urn-I	4	2
Urn-II	2	4

A ball is randomly drawn from urn-I. If it is **red** then put it in urn-II and if it is **white** then put it back in urn-I, with an additional **white** ball from urn-II. Further a ball is randomly drawn from urn-II. If it is **red** then put it back in urn-II, with an additional **red** ball from urn-I and if it is **white** then put it in urn-I. After these pair of operations **same number of white balls were found in urn-I as initially OR same number red balls were found in urn-I as initially**. Let p be the probability that there were same number of red balls as initially in urn-I then $17p$ is

10. Let $P = \frac{(1^4 + \frac{1}{4})(3^4 + \frac{1}{4}) \dots ((2n-1)^4 + \frac{1}{4})}{(2^4 + \frac{1}{4})(4^4 + \frac{1}{4}) \dots ((2n)^4 + \frac{1}{4})}$ and if $\lim_{n \rightarrow \infty} (n^k \cdot P) = L$ if (finite) then value of $\frac{1}{L} - k$ is

Space for Rough Work

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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